

Customer Shopping Behavior Analysis

Comprehensive Technical & Strategic Business Insights Report

1. Executive Overview & Objectives

This project delivers an in-depth empirical analysis of customer shopping behavior using a dataset comprising 3,900 transactional records across diverse consumer retail product categories. The primary objective is to uncover critical insights into customer spending dynamics, demographic segmentation, product category performance, and subscription program engagement to directly inform high-impact strategic business decisions.

Key Business Performance Indicators

Total Customers	Avg Purchase Spend	Avg Review Rating	Subscription Rate
3,900 Records	\$59.76 USD	3.75 / 5.00	27% Active (73% Non)

2. Dataset Architecture & Summary

The underlying dataset provides a multi-dimensional view of retail transactions. It consists of 3,900 rows and 18 attributes, structured into distinct analytical categories:

- Demographic Attributes:** Customer ID, Age, Gender, Geographic Location, and Subscription Status.
- Transactional & Product Details:** Item Purchased, Product Category, Purchase Amount (USD), Season, Size, and Color.
- Behavioral & Logistics Indicators:** Discount Applied status, Promo Code Usage, Previous Purchases count, Purchase Frequency, Review Rating, and Shipping Type.
- Data Quality Assessment:** A total of 37 missing values were identified exclusively within the 'Review Rating' feature.

Exploratory Data Summary Statistics

Metric / Dimension	Age (Years)	Purchase Amount (\$)	Review Rating	Previous Purchases
Count	3,900	3,900	3,863 (Cleaned)	3,900
Mean	44.07	\$59.76	3.75	25.35
Std Deviation	15.21	\$23.69	0.72	14.45

Min	18.00	\$20.00	2.50	1.00
25% (Q1)	31.00	\$39.00	3.10	13.00
50% (Median)	44.00	\$60.00	3.80	25.00
75% (Q3)	57.00	\$81.00	4.40	38.00
Max	70.00	\$100.00	5.00	50.00

3. Exploratory Data Cleaning & Preparation (Python Pipeline)

Data pre-processing was conducted using Python (pandas) to ensure strict data hygiene, consistency, and analytical integrity:

- Data Ingestion & Structural Audit:** Imported full transactional records into Pandas DataFrames. Inspected dataset shape using `df.info()` and extracted distributional parameters via `df.describe()`.
- Intelligent Missing Value Imputation:** Identified 37 missing values in 'Review Rating'. Rather than dropping records, missing values were imputed using category-specific median review ratings to preserve statistical distribution.
- Standardization & Naming Conventions:** Converted all schema column titles into standardized snake_case notation for automated SQL pipeline compatibility.
- Feature Engineering & Categorization:** Engineered binned 'age_group' categories ('Young Adult', 'Middle-aged', 'Adult', 'Senior') and standardized 'purchase_frequency_days'.
- Redundancy Verification & Deduplication:** Cross-evaluated 'discount_applied' and 'promo_code_used'. Verified 100% statistical collinearity/redundancy and dropped 'promo_code_used' to optimize model dimensionality.
- PostgreSQL Database Integration:** Established SQLAlchemy connection with PostgreSQL database to stream clean relational data into target analysis schema.

4. Key Business Findings & SQL Query Results

Structured query analysis in PostgreSQL answered core strategic questions, driving quantifiable insights across revenue, customer behavior, and product performance:

4.1 Revenue Generation by Gender Segment

Male customers represent the largest segment in total overall spending, contributing over twice the gross revenue generated by female customers due to higher transactional volume.

Gender	Total Revenue (USD)
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Male	\$157,890.00
Female	\$75,191.00

4.2 High-Value Discount-Driven Shoppers

A focused SQL query identified 839 unique customers who leveraged promotional discounts yet still achieved purchase values strictly exceeding the overall average spend (\$59.76). This indicates high price elasticity combined with high purchasing capacity.

4.3 Top 5 Rated Product Offerings

Customer satisfaction ratings highlight accessory and footwear lines as highest-performing in terms of perceived quality:

Item Purchased	Average Review Rating
Gloves	3.86 / 5.00
Sandals	3.84 / 5.00
Boots	3.82 / 5.00
Hat	3.80 / 5.00
Skirt	3.78 / 5.00

4.4 Order Values Across Logistics & Shipping Types

Order values across shipping tiers demonstrate that customers utilizing Express Shipping spend slightly more per transaction than standard shoppers, justifying premium fulfillment incentives.

Shipping Type	Average Transaction Value (USD)
Standard Shipping	\$58.46
Express Shipping	\$60.48

4.5 Subscriber vs. Non-Subscriber Financial Comparison

While average order values are comparable (\$59.49 vs \$59.87), non-subscribers current drive 73% of overall sales volume, highlighting a massive opportunity to convert existing non-subscribers into recurring members.

Subscription Status	Total Customers	Avg Spend (USD)	Total Revenue (USD)
Yes (Subscribers)	1,053	\$59.49	\$62,645.00
No (Non-Subscribers)	2,847	\$59.87	\$170,436.00

4.6 Top Discount-Dependent Products

Certain categories show extreme discount dependency, with nearly half of all purchases driven by promotional pricing:

Item Purchased	Discounted Purchase Rate (%)
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

4.7 Behavioral Customer Segmentation Breakdown

Customers were categorized based on historical order frequency into New (1-2 purchases), Returning (3-5 purchases), and Loyal (>5 purchases):

Customer Segment	Number of Customers
Loyal Customers (>5 orders)	3,116
Returning Customers (3-5 orders)	701
New Customers (1-2 orders)	83

4.8 Top Category-Level Best Sellers

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4.9 Revenue Contribution by Age Demographic

Age Group	Total Gross Revenue (USD)
Young Adult	\$62,143.00
Middle-aged	\$59,197.00
Adult	\$55,978.00
Senior	\$55,763.00

5. Power BI Interactive Executive Dashboard

To empower operational executive decision-making, a business intelligence dashboard was architected in Power BI. The layout integrates real-time slicers (Gender, Category, Subscription, Shipping Type) with high-level KPI cards and comparative charts:

- **Executive KPI Summary Cards:** Instant display of overall metrics: 3.9K Total Customers, \$59.76 Average Purchase Value, and 3.75 Overall Average Review Rating.
- **Subscription Penetration Chart:** Donut visualization indicating 27% Subscribed vs 73% Non-Subscribed breakdown.
- **Category Performance Visuals:** Comparative bar charts illustrating top sales volume and revenue generation led by Clothing and Accessories.
- **Demographic Revenue Distribution:** Horizontal breakdown displaying consistent revenue capture across Young Adult, Middle-aged, Adult, and Senior brackets.

6. Strategic Business Recommendations

Based on cross-sectional analysis of customer data, SQL query findings, and Power BI visual trend lines, we propose 5 key strategic growth pillars:

1. **Scale Subscription Program Conversion:** With 73% of customers currently non-subscribers and highly loyal repeat purchase rates (>3,000 loyal customers), implement targeted post-purchase conversion funnels offering exclusive perks (e.g., free express shipping, priority access).
2. **Optimize Promotional & Discount Margins:** Over 49% of purchases in Hats, Sneakers, and Coats are discount-dependent. Implement dynamic minimum spend thresholds (\$75+) to qualify for discounts, preserving profit margins while maintaining high order volumes.

3. Elevate Loyalty Program Engagement: Since over 80% of customer base falls under the 'Loyal' segment (>5 historical purchases), introduce tiered loyalty rewards (Gold/VIP tiers) to incentivize higher basket sizes.

4. Capitalize on High-Performing Categories: Focus marketing ad spend on top-rated items (Gloves, Sandals, Boots) and top revenue generators (Jewelry, Blouses, Jackets) to maximize ROAS.

5. Target Key Demographic & Shipping Channels: Young Adults represent the largest revenue contributor (\$62.1K). Tailor social media marketing and express shipping promotions directly toward this demographic segment.